

Software Requirements Specifications

Project Title: Micro-Service Hub

Part 1: Introduction

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1.Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for the **Micro-Service Hub Platform**. This document serves as the primary technical specification for the project team and course evaluators to guide the development, testing, and assessment of the system.

1.2 Document Conventions

- **Section Structure:** Follows standard software documentation guidelines for clear navigation.
- **Typographical Conventions:** Key system entities, technology names, and specific module features (e.g., **NID Verification**, **Escrow Payment**) are emphasized using bold text.
- **Requirement Standards:** The keywords "**Shall**" (mandatory functionality) and "**Should**" (recommended functionality) indicate requirement priority.

1.3 Intended Audience

This document is intended for:

- **Course Instructors and Evaluators:** For reviewing project scope, architectural feasibility, and academic fulfillment.
- **Project Developers:** For implementing database schemas, backend APIs, and frontend interfaces.
- **Quality Assurance Testers:** For drafting system test cases and validating feature requirements.

1.4 Project Scope

The **Micro-Service Hub Platform** is a web application that connects users needing small, everyday tasks performed with nearby verified service providers.

- **In-Scope Features:**

- User authentication with **NID Verification** for identity validation.
- Task management including posting, applying for, and accepting micro-services.
- Location-based service discovery using **Google Maps / OpenStreetMap**.
- Real-time messaging between users via **Socket.IO**.
- Secure payment simulation using an escrow mechanism with sandbox payment APIs.
- Review, and dispute management handled via an **Admin Dashboard**.
- **Out-of-Scope Features:**
 - Native mobile applications (iOS / Android).
 - Integration with commercial courier or logistics systems.

1.5 References

- **IEEE Std 830-1998:** IEEE Recommended Practice for Software Requirements Specifications.
- **MERN Stack Documentation:** Official documentation for MongoDB, Express.js, React.js, and Node.js.
- **Socket.IO Documentation:** Technical references for real-time event-driven communication.